



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For



AI Security Governance for Businesses

TGS Ref No: TGS-2026061329

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	1 September 2026	Initial release — 2-day WSQ lesson plan for AI Security Governance for Businesses, aligned to the four topics and the 16 hands-on labs, with slide references to deck v1.0.	Dr Alfred Ang

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Course Information

Course Title	AI Security Governance for Businesses
WSQ Course Reference	TGS-2026061329
TSC Title / Code	Responsible AI and Generative AI Practices · ICT-INT-0055-1.1 · Level 1
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 8 training hours per day (16 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led classroom, synchronous Zoom or corporate on-site; hands-on governance labs using browser-based security tools
Trainer	Dr Alfred Ang
Trainer Slides	AI Security Governance for Businesses-v1.0.pptx

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Analyse AI security governance foundations and the business risks that AI, generative AI and agentic AI systems introduce across an organisation.
- LO2: Evaluate AI governance frameworks and build an organisational governance structure of policies, roles, risk ownership and accountability.
- LO3: Develop and apply governance controls across the AI lifecycle, from data sourcing and model development through deployment, monitoring and decommissioning.
- LO4: Implement security controls for agentic AI, conduct an AI risk assessment and produce a prioritised AI security governance implementation roadmap.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Case Study (CS) — one coherent governance case study, 1 hour, open book.
- Format: Open Book — course slides, Learner Guide, the learner's own lab outputs and approved materials only.
- Final assessment is administered on Day 2 after the 8 instructional hours: the Written Assessment (1 hour) followed by the Case Study (1 hour) — 120 minutes in total.
- Grading: Competent (C) / Not Yet Competent (NYC) on each instrument.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Day 1 — Foundations, Business Risks and Building the Governance Framework

Time	Duration	Topic / Activity (with slide references)
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9:30–10:00	30 min	Welcome, course introduction, learning outcomes, ground rules and mandatory digital attendance (AM)
10:00–11:15	75 min	Core concepts: why AI changes the security picture; the expanded attack surface; traditional vs AI-era threats; the 12 NIST GenAI risks; the lethal trifecta (slides 54+)
11:15–11:30	15 min	Tea break
11:30–13:00	90 min	Topic 1 — AI Security Governance Foundations and Business Risks (concepts + demo). Hands-on: Lab 1: Build the AI System Inventory for NovaBank (slide 61); Lab 2: Map the AI Threat Landscape with the Threat Simulator (slide 64)
13:00–14:00	60 min	Lunch break — mandatory digital attendance (PM) on return
14:00–15:30	90 min	Hands-on: Lab 3: Analyse the Lethal Trifecta in a Live Agent Design (slide 67); Lab 4: Build the Business Case for AI Security Governance (slide 70)
15:30–15:45	15 min	Tea break
15:45–18:15	150 min	Topic 2 — Building an AI Governance Framework: NIST AI RMF, MGF for GenAI, policy architecture, roles and RACI (concepts) (slides 74+). Hands-on: Lab 5: Run a NIST AI RMF Gap Assessment (slide 83); Lab 6: Draft the AI Acceptable Use and Governance Policy Set (slide 86); Lab 7: Design the Governance Operating Model and RACI (slide 89); Lab 8: Apply the PDPA and PDPC AI Advisory Guidelines (slide 92)
18:15–18:30	15 min	Day 1 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 2 — Lifecycle Controls, Agentic AI Security and the Implementation Roadmap

Time	Duration	Topic / Activity (with slide references)
9:30–9:45	15 min	Day 1 recap and mandatory digital attendance (AM)
9:45–11:15	90 min	Topic 3 — Governance Controls Across the AI Lifecycle: data governance,

		secure development, testing and red-teaming, deployment gates, monitoring, incident response, decommissioning (concepts + demo) (slides 96+)
11:15–11:30	15 min	Tea break
11:30–13:00	90 min	Hands-on: Lab 9: Govern the Data: Classification, Minimisation and Protection (slide 107); Lab 10: Red-Team an AI System Before Deployment (slide 110)
13:00–14:00	60 min	Lunch break — mandatory digital attendance (PM) on return
14:00–15:15	75 min	Hands-on: Lab 11: Design Deployment Gates and Continuous Monitoring (slide 113); Lab 12: Run an AI Incident Response Exercise (slide 116)
15:15–15:30	15 min	Tea break
15:30–17:15	105 min	Topic 4 — Security Controls for Agentic AI, Risk Assessment and Implementation Roadmap: agentic threat model, identity and least agency, guardrails, permission ladders, CSA agentic RMF profile (concepts) (slides 120+). Hands-on: Lab 13: Design the Agentic Security Control Stack (slide 133); Lab 14: Apply the CSA Agentic Extensions to the NIST AI RMF (slide 136); Lab 15: Conduct a Full AI Security Risk Assessment (slide 139); Lab 16: Capstone — Build the AI Security Governance Roadmap (slide 142)
17:15–18:30	75 min	Course recap and consolidation of the capstone roadmap; Q&A; TRAQOM survey; Briefing for Assessment; Assessment digital attendance

Total training time: 480 minutes (8 hours).

Assessment Administration (Day 2, after the instructional hours)

Time	Duration	Assessment Instrument
18:30–19:30	60 min	Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book
19:30–20:30	60 min	Case Study (CS) — one coherent governance case study, 1 hour, open book

Total assessment time: 120 minutes (2 hours), additional to the 8 instructional hours above. Both instruments are open book. The Assessment digital attendance is taken before the papers are issued. Where the schedule requires, the assessment may be administered on a separate session; the instruments and their durations do not change.

Lab Reference (aligned to topics and slide numbers)

Topic	Weighting	Labs	Deck slides
Topic 01: AI Security Governance Foundations and Business Risks	25%	Lab 1, Lab 2, Lab 3, Lab 4	54 · labs 61–70
Topic 02: Building an AI Governance Framework: Policies, Roles and Risk Ownership	25%	Lab 5, Lab 6, Lab 7, Lab 8	74 · labs 83–92
Topic 03: Governance Controls Across the AI Lifecycle	25%	Lab 9, Lab 10, Lab 11, Lab 12	96 · labs 107–116
Topic 04: Security Controls for Agentic AI, Risk Assessment and Implementation Roadmap	25%	Lab 13, Lab 14, Lab 15, Lab 16	120 · labs 133–142

Lab Detail

Lab	Title	Learners produce	Slide
1	Build the AI System Inventory for NovaBank	A completed AI System Inventory (ai-inventory.csv) covering 12 AI systems with owner, purpose, data classes, model, autonomy level and risk tier, plus a shadow-AI findings note.	61
2	Map the AI Threat Landscape with the Threat Simulator	A completed Threat-to-Business-Impact map covering 8 threat classes, each with its AI-era variant, the NovaBank asset at risk and a first-line control.	64
3	Analyse the Lethal Trifecta in a Live Agent Design	A completed agent risk analysis: the trifecta assessment, five mapped NIST GenAI risks, a documented attack path and a redesign that breaks the trifecta.	67
4	Build the Business Case for AI Security Governance	A one-page board paper with a quantified exposure estimate, three prioritised investments and the expected risk	70

		reduction for each.	
5	Run a NIST AI RMF Gap Assessment	A completed NIST AI RMF gap assessment across 16 subcategories, each scored with evidence, plus a top-10 prioritised gap register.	83
6	Draft the AI Acceptable Use and Governance Policy Set	A complete AI Acceptable Use Policy (2 pages) plus a one-page AI Risk Management Standard, both mapped to MGF dimensions and RMF subcategories.	86
7	Design the Governance Operating Model and RACI	A governance operating model diagram, committee terms of reference, six role descriptions and a RACI matrix covering twelve AI governance controls.	89
8	Apply the PDPA and PDPC AI Advisory Guidelines	A PDPA compliance determination for four AI use cases, each with lawful basis, required notification, accountability measures and documentation.	92
9	Govern the Data: Classification, Minimisation and Protection	A governed training dataset specification: a field-by-field classification and minimisation decision, a de-identification method per field, and a tested hashing approach with retention rules.	107
10	Red-Team an AI System Before Deployment	A pre-deployment red-team report: findings with severity, evidence and OWASP mapping, extended with the AI-specific test cases and a go/no-go recommendation.	110
11	Design Deployment Gates and Continuous Monitoring	A deployment gate checklist with named approvers, plus a monitoring specification with ten metrics, thresholds, alert routing and response actions.	113
12	Run an AI Incident Response Exercise	A completed incident record: timeline, containment decision, personal-data impact assessment, PDPA notification determination and five improvement actions.	116
13	Design the Agentic Security Control Stack	A layered agentic control specification: agent identity and scopes, guardrail rules, a deny/ask/allow	133

		permission policy for 12 tools, sandbox boundaries and human approval gates.	
14	Apply the CSA Agentic Extensions to the NIST AI RMF	An agentic governance pack: five agents tiered with justification, a delegation accountability record, a completed agent registry and a behavioural telemetry specification.	136
15	Conduct a Full AI Security Risk Assessment	A complete AI security risk assessment: 12 rated risks with likelihood, impact, existing controls, residual rating, treatment decision, owner and target date.	139
16	Capstone — Build the AI Security Governance Roadmap	A 12-month AI security governance roadmap across three phases with initiatives, owners, effort, dependencies and success metrics, plus a five-slide executive presentation.	142

Training Resources

- Trainer Slides (PPT) and Learner Slides (PDF) — AI Security Governance for Businesses-v1.0
- Learner Guide (DOCX/PDF) with the full step-by-step procedure for all 16 labs
- Lab pack — one folder per lab with mock data and a printable instruction PDF
- Browser-based lab tools (no installation required):
 - Cybersecurity Threat Simulator — <https://alfredang.github.io/cybersecuritysimulator/>
 - Hacklab — Ethical Hacking Lab Simulator — <https://alfredang.github.io/ethnicalhacking/>
 - FauxBank — Pentest Training Sandbox — <https://pentest-fauxbank.vercel.app/>
 - Cryptography Toolkit — <https://alfredang.github.io/cryptography-toolkit/>
- Whiteboard/flipchart for the incident-response tabletop exercise (Lab 12)
- LMS/TMS portal for course materials and assessment submission — <https://lms-tms.tertiaryinfotech.com>